



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Pilani Campus

AUGS / AGSR Division

FIRST SEMESTER 2026-2027

COURSE HANDOUT(PART-II)

Date: 7/31/2026

In addition to Part-I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No.	: CS F222
Course Title	: Discrete Structures for Computer Science
Instructor In Charge	: L. Rajya Lakshmi
Instructor	: L. Rajya Lakshmi (rajya.lakshmi@pilani.bits-pilani.ac.in) : Sundaresan Raman (sundaresan.raman@pilani.bits-pilani.ac.in) : Poonam Goyal (poonam@pilani.bits-pilani.ac.in) : Nikhil Maurya (p20230031@pilani.bits-pilani.ac.in) : Prashant Umrao (p20240433@pilani.bits-pilani.ac.in)

1. COURSE DESCRIPTION

This course aims in providing mathematical foundations for many other computer science courses including data structures and algorithms, automata, compiler theory, databases, operating system etc. This course encompasses topics including sets & operation on sets; relations & equivalence relations; number theory; weak & strong form of mathematical induction; principle of inclusion & exclusion, pigeonhole principle; recurrence relations; digraphs & graphs, graph isomorphism & sub-graphs, spanning trees, Euler & Hamiltonian graphs, planar graphs, chromatic numbers & graph coloring; groups.

2. SCOPE & OBJECTIVE:

To develop logical and mathematical concepts necessary to appreciate computational systems. & study of concepts, techniques, and skills necessary to comprehend the structure of problems encountered in design and analysis of algorithms. To provide mathematical foundations for courses in computer science that rely upon the comprehension of formal abstract concepts. To study recursion and to write recursive definitions for certain sequences and collections of objects. Graphs, directed graphs, planar graphs & their relevance to circuit design & map coloring problems. Trees and their applications. Basic algebraic structures and their applications.

3. TEXT BOOK:

Mott, Kandel, & Baker: Discrete Mathematics for Computer Scientists & Mathematicians , PHI, 2e, 2002.:

4. REFERENCE BOOKS:

R1. K H Rosen: Discrete Mathematics & its Applications, TMH, 7e, 2011.

R2. Douglas West: Introduction to Graph Theory, Pearson, 2e, 2001.

R3. GRAPH THEORY WITH APPLICATIONS, J. A. Bondy and U. S. R. Murty

5. COURSE PLAN

Module No.	Lecture Session	Reference	Learning outcomes
Module 1: Course Introduction and Background	L1-L2 Course Introduction, Importance of Discrete Structures	Lecture Slides, Ch. 1	Course Introduction, Basics of Set Theory, Importance of studying Discrete structures in computer science,
Module 2: Induction and Recursion	L3-L7	Ch.1 (1.8, 1.10), R1: Ch. 4.1, Ch. 4.2, R1: Ch. 4 (4.3, 4.4)	Methods of Proving, Mathematical Induction, Using mathematical induction to prove the theorems or statements, Strong induction and well ordering principles, Recursive or Inductive definitions of functions, sets, and structures, Structural induction, proofs using structural induction, Problem solving using Recursion, Recursive Algorithms, Recursion vs Iteration
Module 3 Relations	L8-L18 Properties of relations, representing relations, closure of relations, Equivalence Classes, Partial Ordering, Lattices	Ch.4 (4.1, 4.2, 4.5), R1: Ch. 7 (7.1, 7.2), Ch. 4.7, R1: Ch. 7.3, Ch. 4.6, R1: Ch. 7.4, Ch. 4 (4.3, 4.4), R1: Ch. 7.3	Mathematical definitions of relations, n-ary relations, Properties: reflexive, symmetric, transitive, antisymmetric, and asymmetric, Operations on Relations, Matrices, Diagraphs, Reflexive, Symmetric, and Transitive closures, Warshall's Algorithm, Equivalence Relations and Classes, Partial Ordering Relations, lattices, Hasse Diagram
Module 4 Graph Theory – Part1	L19-L27 Introduction to Graph Theory, Graph Isomorphism and Connectivity, Euler and Hamiltonian Graphs	Ch. 5.1, R1: Ch.8 (8.1, 8.2, 8.3), Ch. 5 (5.1, 5.2), R1: Ch. 8 (8.3, 8.4), Ch. 5 (5.9, 5.10), R1: Ch. 8.5	Basic Concepts, Special Graphs, Elementary Theorems on Graphs, Graph Representation: Adjacency & Incidence Matrices Problem modeling, Path and Cycles, Isomorphism in Graphs, Connected graphs, connected components, cut-vertices, and cut-edges, Applications, Definitions, Conditions (necessary and sufficient) for existence of Euler path and Hamiltonian cycles
Module 5 Graph Theory – Part2	L28-236 Planar Graphs and Graph Coloring, Matching, Graph Algorithms	Ch. 5 (5.7, 5.11), R1: Ch. 8 (8.7, 8.8), R2. Ch. 3, R1: Ch. 8.6	Planar Graphs, Euler formula for planer graphs, characterization of planar graphs, Graph Coloring: Vertex and Edge coloring, Chromatic graphs chromatic number. Basic bounds on chromatic number. Five color theorem, Applications of Graph planarity, Applications of graph coloring, Matching and Covering in Bi-partite graph, Perfect Matching, Maximal Matching and Augmented Paths, Applications, Graph Traversal: BFS, DFS, Spanning Tree: Prim, Kruskal, Shortest Path: Dijkstra, Bellman Ford

Module 6 Trees	L37-39 Introduction, Spanning Trees, Tree Traversal	Ch. 5.3, R1: Ch. 9.1, Ch. 5.4, R1: Ch. 9.4	Introduction, Application, Terminology and Theorems, Tree Traversal Algorithms, Spanning and Minimum Spanning Tree
Module 7 Algebraic Structures	L40-L42 Introduction to Algebraic Structures	R1: Ch. 11 (11.1, 11.2, 11.3, 11.4, 11.5)	The structure of algebra, Types of algebraic systems: Semi-group, monoid, group, generator of a group, Rings, Fields, & Vector Spaces

6. EVALUATION SCHEME:

Component	Duration	Weightage (%)	Date & Time	Remarks
Midsem Test	90 mins.	25	05/10/2026 (AN1)	Closed Book
Quizzes	30-40 mins each	30	TBD	Closed Book
Comprehensive	3 hours	35	3/12/2026 (FN)	Partly Open
Participation linked assessments	In-class	10 (Lectures:6, Tutorials:4)	Lecture/Tutorial	Closed book

Please note if the date/time for an assessment requires changes, it would be announced by IC / AUGSD.

7. CHAMBER CONSULTATION HOUR: Monday: 12 noon

To contact the tutorial instructor, you may contact them via email.

8. MAKE-UP Policy: Make-ups are generally considered only for medical emergencies leading to hospitalization (or a personal emergency of similar nature).

For a foreseen absence, make-up request should be made personally by meeting the Instructor-in-Charge. Reasons for unanticipated absence that qualify one for make-up include medical or similar personal emergencies only. In such cases, the Instructor-in-Charge must be informed either by email or telephone, at the earliest.

Please note that there would be no exclusive makeup for a quiz. There would be one common makeup for all the three quizzes, towards the end of the semester. The decision by the Instructor-in-Charge regarding granting makeups shall be final.

9. ATTENDANCE: Students are expected to attend all the classes & tutorials, and take part in at least 50% of the participation linked assessments, in both lecture as well as tutorials.

10. NC Criteria: Students scoring less than 10% of the marks would be awarded an NC report

11. ANNOUNCEMENTS & NOTICES: Announcements would be made in class and may be followed up with an email. For sharing course material, either Nalanda (nalanda.bits-pilani.ac.in) or Google Drive may be used.

Instructor-in-charge